

torch.Tensor.index_add_#

https://pytorch.org/docs/stable/generated/torch.Tensor.index_add_.html#torch

Description:

Accumulate the elements of alpha times source into the self tensor by adding to the indices in the order given in index. For example, if `dim == 0`, `index[i] == j`, and `alpha=-1`, then the *i*th row of source is subtracted from the *j*th row of self. The *dim*th dimension of source must have the same size as the length of index (which must be a vector), and all other dimensions must match self, or an error will be raised. For a 3-D tensor the output is given as: This operation may behave nondeterministically when given tensors on a CUDA device. See Reproducibility for more information.

Function Signature

`Tensor.index_add_(dim, index, source, *, alpha=1) → Tensor#`

Parameters

Keyword Arguments

Parameters

Keyword

alpha (Number) – the scalar multiplier for source

Code Examples

```
self[index[i], :, :] += alpha * src[i, :, :] # if dim == 0
self[:, index[i], :] += alpha * src[:, i, :] # if dim == 1
self[:, :, index[i]] += alpha * src[:, :, i] # if dim == 2
```

```
>>> x = torch.ones(5, 3)
>>> t = torch.tensor([[1, 2, 3], [4, 5, 6], [7, 8, 9]],
dtype=torch.float)
>>> index = torch.tensor([0, 4, 2])
>>> x.index_add_(0, index, t)
tensor([[ 2.,  3.,  4.],
        [ 1.,  1.,  1.],
        [ 8.,  9., 10.],
        [ 1.,  1.,  1.],
        [ 5.,  6.,  7.]])
>>> x.index_add_(0, index, t, alpha=-1)
tensor([[ 1.,  1.,  1.],
        [ 1.,  1.,  1.],
        [ 1.,  1.,  1.],
        [ 1.,  1.,  1.],
        [ 1.,  1.,  1.]])
```

Notes & Warnings

NOTE

Note This operation may behave nondeterministically when given tensors on a CUDA device. See Reproducibility for more information.

Detailed Documentation

Documentation

Accumulate the elements of α times source into the self tensor by adding to the indices in the order given in index. For example, if $\text{dim} == 0$, $\text{index}[i] == j$, and $\alpha == -1$, then the i th row of source is subtracted from the j th row of self.

The dim th dimension of source must have the same size as the length of index (which must be a vector), and all other dimensions must match self, or an error will be raised.

For a 3-D tensor the output is given as:

This operation may behave nondeterministically when given tensors on a CUDA device. See Reproducibility for more information.

`dim` (int) – dimension along which to index

`index` (Tensor) – indices of source to select from, should have dtype either `torch.int64` or `torch.int32`

`source` (Tensor) – the tensor containing values to add

`alpha` (Number) – the scalar multiplier for source